response, (3) use a fancy *multirate* technique described later in this chapter, or (4) make the correction in software before the DAC (see Chapter 24).

Before leaving this section on sampling, we need to dispel a common myth about analog versus digital signals. As this chapter has shown, the amount of information carried in a digital signal is limited in two ways: First, the number of bits per sample limits the resolution of the *dependent* variable. That is, small changes in the signal's amplitude may be lost in the quantization noise. Second, the sampling rate limits the resolution of the *independent variable*, i.e., closely spaced events in the analog signal may be lost between the samples. This is another way of saying that frequencies above one-half the sampling rate are lost.

Here is the myth: "Since analog signals use continuous parameters, they have infinitely good resolution in both the independent and the dependent variables." Not true! Analog signals are limited by the same two problems as digital signals: *noise* and *bandwidth* (the highest frequency allowed in the signal). The noise in an analog signal limits the measurement of the waveform's amplitude, just as quantization noise does in a digital signal. Likewise, the ability to separate closely spaced events in an analog signal depends on the highest frequency allowed in the waveform. To understand this, imagine an analog signal containing two closely spaced pulses. If we place the signal through a low-pass filter (removing the high frequencies), the pulses will blur into a single blob. For instance, an analog signal formed from frequencies between DC and 10 kHz will have *exactly* the same resolution as a digital signal sampled at 20 kHz. It must, since the sampling theorem guarantees that the two contain the same information.

Analog Filters for Data Conversion

Figure 3-7 shows a block diagram of a DSP system, *as the sampling theorem dictates it should be*. Before encountering the analog-to-digital converter,

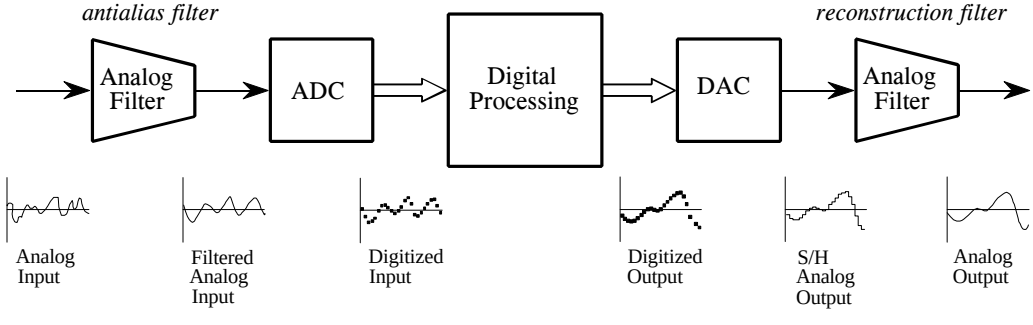

FIGURE 3-7
Analog electronic filters used to comply with the sampling theorem. The electronic filter placed before an ADC is called an *antialias filter*. It is used to remove frequency components above one-half of the sampling rate that would alias during the sampling. The electronic filter placed after a DAC is called a *reconstruction filter*. It also eliminates frequencies above the Nyquist rate, and may include a correction for the zeroth-order hold.